

Effects of hillslope position on soil water infiltration and preferential flow in tropical forest in southwest China

Chunfeng Chen^{a,b,***}, Xin Zou^{a,c}, Ashutosh Kumar Singh^a, Xiai Zhu^{a,b}, Wanjun Zhang^{a,c}, Bin Yang^{a,b}, Xiaojin Jiang^{a,b,**}, Wenjie Liu^{a,b,*}

^a CAS Key Laboratory of Tropical Forest Ecology, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, Menglun, Yunnan, 666303, China

^b Center of Plant Ecology, Core Botanical Gardens, Chinese Academy of Sciences, Menglun, Yunnan, 666303, China

^c University of Chinese Academy of Sciences, Beijing, 100049, China

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ABSTRACT

The hillslope is an essential natural spatial gradient that influences hydrological processes by affecting water distribution, surface flow, soil erosion, and groundwater recharge. To date, few studies have addressed only the hydrological processes of tropical forest hillslopes. To reveal the effect of hillslope on soil hydrological functioning—including water distribution and exchange, infiltration capacity, and flow behaviour—we conducted 36 field infiltration and nine dye-tracer investigations of different hillslope locations in the natural rainforest of Xishuangbanna, southwest China. The soil physical properties—including soil noncapillary and total porosity, saturated water capacity, and field water capacity—decreased with decreasing elevation from hilltop to middle slope and the valley bottom. The water infiltration capacity—including the initial infiltration rate, saturated soil hydraulic conductivity, and average infiltration rate—decreased from the hilltop to the valley bottom. Preferential flow dominated soil water movement more in the upper locations than in the valley bottom. The infiltration capacity parameters and preferential flow were significantly correlated with soil water content, noncapillary and total porosity, root biomass, and termite holes. These results indicated that along with the soil physical properties, root systems, animal activity, cracks, and stones affected the soil infiltration capacity and preferential flow. Differences in the hydraulic processes of each hillslope position contributed to the redistribution, transportation, and storage of surface and belowground water, resulting in differing availabilities of soil water resources and utilisation by plants. The findings of this study can help understand eco-hydrological processes in the context of water resources management in tropical mountain ecosystems.

1. Introduction

Increasing soil degradation due to hillslope erosion, land use change, water depletion, excessive cultivation, and extreme rainfall events is a serious problem, expected to worsen in mountainous regions worldwide in the coming decades (Sweeney et al., 2015; Sun et al., 2019). As fundamental units of hydrological processes, hillslopes are widely involved in their effects on soil erosion and runoff (Hancock et al., 2010), streamflow generation (Londoño et al., 2016; Wu et al., 2018),

water supplementation, and sediment and nutrient transportation (Kouznetsov et al., 2007; Blake et al., 2010), and hillslope stability (Guo et al., 2018, 2019). In particular, soil infiltration—a function of preferential water flow, saturated hydraulic conductivity (K_s), and soil structures—plays an essential role in hydrological processes of mountain ecosystems. Infiltration is a key factor that controls runoff generation, groundwater recharge, water exchange, and eventually vegetation growth and development (Levy et al., 2011; Benegas et al., 2014). Many studies have assessed the preferential flow, K_s , and soil structures of

* Corresponding author. Key Laboratory of Tropical Forest Ecology, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, Menglun, Mengla, Yunnan, 666303, China.

** Corresponding author. CAS Key Laboratory of Tropical Forest Ecology, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, Menglun, Yunnan, 666303, China.

*** Corresponding author. CAS Key Laboratory of Tropical Forest Ecology, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, Menglun, Yunnan, 666303, China.

E-mail addresses: jiangxiaojin@xtbg.ac.cn (X. Jiang), lwj@xtbg.org.cn (W. Liu).

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hillslopes (Mei et al., 2018; Zhao et al., 2020); however, their combined influence on the hydrological processes in mountainous regions has received limited attention, especially in the tropical mountain ecosystems.

Water infiltration can be divided into matrix flow and preferential flow, based on the soil water transportation path. Matrix infiltration refers to a comparatively slow water (and solute) movement, in which water permeates through the soil body, mainly driven by soil capillary action (Zhang et al., 2017). Macropore flow commonly occurs when water (and solute) flow through soil macropores and largely bypass the soil matrix (Jiang et al., 2017). Subsurface lateral flow refers to processes of water infiltrating laterally downslope within a permeable soil layer but along the upper surface of a less permeable layer (Gerke et al., 2015). Field dye-tracer experiments and observations in hillslope soils have demonstrated that the prevalence of preferential flow processes was affected by various complicating factors and prominently varied across different environments and vegetation types (Jost et al., 2012; Mei et al., 2018). For example, Jost et al. (2012) found that soil in beech hillslopes had faster water flow and a higher preferential flow than in spruce hillslope soils because of the higher number of macropores formed by coarse and old roots. In boreal forest hillslopes, Laine-Kaulio et al. (2015) found that lateral preferential flow dominated the mid-slopes, whereas macropore flow was prevalent at both topslopes and downslopes, resulting from the different rock depths and soil layers. However, Mei et al. (2018) found that macropore flow was prevalent in the upslope and midslope because of the greater root network, while lateral flow dominated the downslope owing to the coverage of rock fragments. Soil structures (e.g., soil porosity, macropore networks, heterogeneity, and anisotropy) and animal activities (termite holes and earthworm burrows) vary with different hillslopes, creating preferential flow paths—or allow water to create preferential paths—and promote the hydraulic conductivity of the soil (Jiang et al., 2017; Marquart et al., 2020).

K_s is an important hydraulic parameter of water and solute transportation models, affecting water storage and movement in soils (Gwenzi et al., 2011). Higher K_s can promote rainwater infiltration and reduce runoff generation and hillslope erosion. In hillslope situations, K_s shows significant spatial variability due to various soil properties, such as sand content, agricultural activities, and bulk density (Papanicolaou et al., 2015). In addition, the soil K_s is affected by rock fragments, vegetation cover, root network, soil fauna, and microorganisms, all of which vary across different hillslope positions. K_s and preferential flow paths characterise the water movement and facilitate hydraulic redistribution in soil profiles (Jiang et al., 2017). Runoff-generating interactions between precipitation and geomorphological features determine hillslope infiltration and flow paths, increasing spatial and temporal variability in mountainous-region water supplies. Previous studies have found higher water transportation, redistribution, and soil storage associated with soils exhibiting higher preferential flow, allowing the plants to acquire different water sources and promote water use efficiency (Jiang et al., 2017; Zhu et al., 2019). Most studies support the idea that infiltration and preferential flow can diversify the water supply to plants; however, only a few studies have investigated the effect of hydrological characteristics on the soil water supply mechanism at various hillslope locations.

The effect of hillslope location on infiltration rate and preferential flow has been studied in the mountainous regions of the karst (Chen et al., 2009), loess (Mei et al., 2019), and boreal forest (Laine-Kaulio et al., 2015). However, studies are notably lacking in the humid tropical forests region, particularly concerning the effect of hillslope location and orientation on soil hydrological functions and services, such as water infiltration, storage, and regulation. Therefore, this study was conducted in the Xishuangbanna (SW China), having a typical humid tropical climate characterised by accelerated surface runoff and soil erosion in the rainy season and narrow streamflow and water storage capacity in the dry season. The objectives of the present study were to

(1) reveal the effect of slope position on infiltration capacity, preferential flow, and soil structures, and explore the related influencing factors; and (2) identify the water supply mechanisms and ascertain hydrological characteristics of different slope positions, and thereby promote hydrological management in tropical mountainous areas. We hypothesised that (i) soil infiltrability and preferential water flow would vary across different slope positions; (ii) varying hydrological processes across different slope positions were closely associated with the different soil structures, vegetation coverage, and animal activities; and (iii) different slope positions had different functions in hydrological processes and services in tropical mountain ecosystems. The findings of this study can help understand the hydrological processes of hillslopes for watershed management of the humid tropical region.

2. Materials and methods

2.1. Study site and experimental design

The study was conducted in the Xishuangbanna Tropical Botanical Garden (21°55'39" N, 101°15'55" E) in Menglun, Mengla, southwestern China. The region belongs to a mountainous zone, the southern extension of the Hengduan Mountains, and approximately 95% is covered by mountains and hills. The mean annual precipitation is ~1377 mm (Fig. 1), approximately 85% of which occurs during the rainy season. The annual mean air temperature is 21.6 °C. The potential evapotranspiration (determined by evaporation pans) varies between 1000 and 1200 mm per year.

The study was conducted on hillslopes of mountains which are occupied by tropical rainforests (Fig. S1). Nine 20 × 20 m plots were randomly selected by GPS coordinates on southwest-facing sections of hillslope. Three replicated plots were established at the hilltop (elevation 610.5 m), three on the middle slope (elevation 565.2 m), and three at the valley bottom (elevation 528.0 m), the locations of each elevation were separated by ~350 m. At each elevation, the three plots were separated by ~50 m. The hilltops had a slope of 10–15° and dense vegetation growth. The conspicuous tree layer was dominated by *Lagerstroemia villosa*, *Castanopsis indica*, and *Pittosporopsis kerrii*; the shrub layer mainly included *Ficus hirta* and *Psychotria yunnanensis*; and the herb layer is dominated by *Didymosperma caudatum*, *Adiantum capillus-veneris*, and *Bolbitis heteroclita*. The middle slope had slopes of 12–18° and contained a tree layer (*F. racemosa*, *Pterocarya stenoptera*, *Phoebe lanceolata*, *Albizia lucidior*), a shrub layer (*Murraya exotica* and *Ardisia brunneascens*), and a herb layer (*Piper longum*, *Alocasia macrorrhiza*, and *Ophiopogon bodinieri*). The valley bottom had a slope of 0–5° with a tree layer (*Baccaurea ramiflora*, *L. villosa*, *Millettia pulchra*, *Pometia pinnata*, and *P. kerrii*), a simple shrub layer (*Harpullia cupanioides* and *Pavetta hongkongensis*), and a herb layer (*B. heteroclita*, *Phrynium capitatum*, *Pteridrys cnemidaria*, and *Elatostema rupestre*).

2.2. Saturated hydraulic conductivity (K_s)

The measures of field K_s were conducted following the procedures described by Zhu et al. (2019). Specifically, a stainless-steel cylinder (diameter 10.0 cm and height 15.0 cm) was inserted 5 cm into the soil after removing the leaf litter, ensuring the sidewalls and edges were watertight. The 10 cm of the cylinder above the soil level was filled with tap water and observed against a reference ruler. The decline in water levels was recorded at 2 min intervals for 20 min, and at 4 min intervals thereafter. The initial infiltration rate (IIR) was calculated as the decrease in depth (cm) over the initial infiltration time (2 min). The average infiltration rate (AIR) was calculated as the total decline in water depth over the total infiltration time (the time for all the water in the cylinder to pass into the soil). The steady-state infiltration rate (I_s , cm min⁻¹) was calculated from the last five recorded values, which consistently demonstrated a steady state of flow; this was after approximately 1.5 h. The I_s was transformed to a quasi-steady

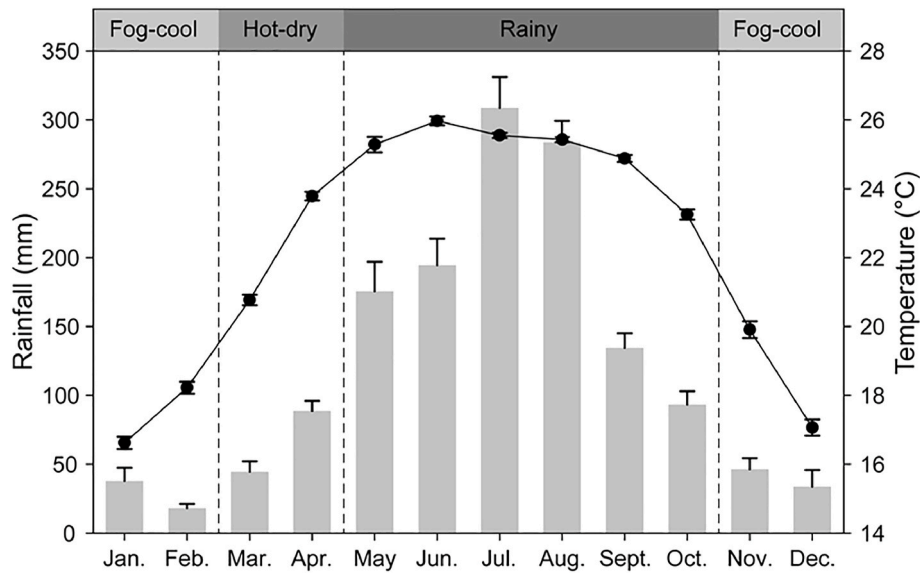


Fig. 1. Mean monthly precipitation and temperature in the last 20 years (2000–2019). Weather data were obtained from the Xishuangbanna Station for Tropical Rainforest Ecosystem located near the study sites. The solid line indicates the monthly mean temperature, and the bar chart represents monthly precipitation.

infiltration rate (SIR , cm min^{-1}) at a standardised 10°C equivalent, in accordance with Eq. (1). The field hydraulic conductivity of saturated soil (K_s , cm min^{-1}) was calculated according to Eq. (2) (Youngs et al., 1993):

$$SIR = \frac{I_s}{0.7 + 0.03T} \quad (1)$$

$$K_s = \frac{SIR}{\frac{H_s}{C_1 d_s + C_2 r_s} + \frac{1}{S(C_1 d_s + C_2 r_s)} + 1} \quad (2)$$

where, I_s is the steady-state infiltration rate under the actual temperature and SIR is the quasi-steady infiltration rate at 10°C . C_1 and C_2 are dimensionless constants with values of 0.316π and 0.184π , respectively. H_s is the height of the water level decline—the difference between the highest and lowest water levels—and was fixed at 5 cm. d_s is the depth of the cylinder inserted into the soil, r_s is the radius of the cylinder, and S is the macroscopic capillary length of the soil. The values of d_s , r_s , and S are 5 cm, 5 cm, and 0.12 cm^{-1} , respectively.

2.3. Dye-tracer infiltration experiment

Nine dye-tracer experiments were performed at the hilltop, middle slope, and valley bottom locations to determine the water infiltration and to visualise the water flow pathways. A hollow stainless-steel cylinder (diameter 20.0 cm and height 30.0 cm) was inserted 5 cm into the soil, ensuring that the sidewalls and edges were watertight. A total of 25 L of 4.0 g L^{-1} Brilliant Blue FCF dye solution was poured slowly into the cylinder, and the total time for all the water to infiltrate the soil was recorded. Then, the cylinders and plots were covered with plastic sheets to prevent evaporation. The sheets and cylinders were gently removed 24 h after the end of the infiltration. Vertical soil sections (100 cm width and 120 cm depth) were then neatly excavated from the centre of the cylinder position, ensuring all the dyed areas were included in the section. The dye-stained patterns of the sections, with a calibrated steel tape placed along the horizontal and vertical axes of the profile for reference, were recorded and photographed using a Canon EOS 6D Mark II camera. In addition, the numbers of the holes, including visible holes created by termites and ants, were counted in the sections with 30 cm depth and 100 cm width (Léonard and Rajot, 2001).

According to the image analysis procedure described by Cey and Rudolph (2009) and Zhu et al. (2019), the profile images were classified

into non-dyed and dyed areas—and the latter then divided into three relative classes (representing the quantity of dye absorbed by the soil, namely 0.05–0.5, 0.5–2.0, and $>2.0 \text{ g L}^{-1}$) according to the dye-staining concentration categories—using ERDAS IMAGINE 9.2 (Erdas Inc., Atlanta, Georgia, USA) with the Supervised Classification tool. The amount of water residue after infiltration, and the water flow patterns, were represented by the Brilliant Blue FCF dye concentration in the images (Jiang et al., 2019). The dyed-stained areas with a higher concentration ($>2.0 \text{ g L}^{-1}$) indicated active water areas, which were dominated by preferential water flow. The dye-stained regions with a lower concentration ($<2.0 \text{ g L}^{-1}$) were defined as the water buffer zones, where water exchange occurred between preferential flow and the soil matrix. The areas of different dye concentration categories were calculated using ArcGIS 10.3 (ESRI Inc., Redlands, CA, USA) mapping software. This geospatial data analyzing tool was used to process dye-stained images to compute the area occupied by each class within different zones, defined by a polygon shapefile. The polygon shapefile was split into a grid having 60 rectangular 160 cm^2 polygons ($80 \text{ cm wide} \times 2 \text{ cm high}$).

In addition, a $20 \times 120 \text{ cm}$ dye-stained image, decoupled from the total dye-stained image, was analysed using ArcGIS 10.3 to obtain the parameters as indices of preferential flow (Bargués Tobella et al., 2014; Fig. S2). The indices of preferential flow were calculated using the following parameters:

Dye coverage (DC), the proportion of the dye-stained area to the total area, was calculated according to Eq. (3).

$$DC = \frac{DS}{DS + ND} \times 100\%, \quad (3)$$

where, DC (%) is the dye coverage, DS is the dye-stained region (cm^2), and ND is the non-stained region (cm^2).

Uniform infiltration depth ($UniFr$) refers to the depth at which DC was reduced below 80%.

Preferential flow fraction ($PF-fr$) is the fraction of total infiltration that flows through preferential flow paths:

$$PF-fr = 100 \times \left(1 - \frac{UniFr \times W}{TotStAr} \right), \quad (4)$$

where, $PF-fr$ is the preferential flow fraction (%), $UniFr$ is the uniform infiltration depth (cm), and $TotStAr$ is the total stained area (cm^2). W is the soil profile width (20 cm in this study).

Length index (*LI*) is the sum of the absolute differences between the consequent dye coverage values with depth in a profile (Bargués et al., 2014):

$$LI = \sum_{i=1}^{60} |DC_{i+1} - DC_i|, \quad (5)$$

where, *LI* is the length index, and *i* represents a given depth interval of the rectangular area in which the dye coverage was calculated.

Peak index (*PI*) is defined as the number of times the vertical line of the total dye coverage intersected the dye coverage profile. The *LI* and *PI* values reflect the heterogeneity of the stained patterns in the soil profiles; a higher value suggesting a higher degree of preferential flow.

2.4. Root biomass calculation

Root biomass was measured by conversion using the density value obtained from a 10 cm soil corer. After the dye tracer experiments, the roots were sampled in the same vertical soil sections (see Fig. 1S). Specifically, soil cores were collected together with the profiles at depths of 0–20, 20–40, 40–60, 60–80, 80–100, and 100–120 cm. All samples were transported to the laboratory, and the roots were collected after manual separation with an elutriator. Living roots were immediately separated by diameter into fine roots ($d < 2$ mm) and coarse roots ($d > 2$ mm). All fresh samples were oven-dried at 65 °C until constant root weights (kg) were obtained and the root biomass (kg m^{-2}) was calculated.

2.5. Soil physical characteristics

Soil physical properties were measured for undisturbed soil cores using the cutting ring method at depths of 0–20, 20–40, 40–60, 60–80, 80–100, and 100–120 cm. The samples were collected in four repetitions using cutting cylinders of known weight (W_{wcc}) with a 50.0 mm inner diameter and a 51.0 mm height in each layer. The cutting cylinders with the samples were transported immediately to the laboratory, weighed (W_{ccs} , g), and placed on the trays filled with distilled water. The water was filled to the cylinder's top edge to ensure that there was no spillover from the top. The cylinders with water-saturated soil samples (W_{sat} , g) were weighed after 24 h of water absorption. Then, the cylinders were laid on salvers with a dry sand layer on the bottom, and their weights were recorded after the water in the samples was drained by gravity for 2 h (W_{ws2h} , g) and 5 d (W_{ws5d} , g), respectively. The indices of the soil physical properties were calculated using the following formulae:

$$\text{Bulk density}(\text{gcm}^{-3}) = \frac{W_{ccs}(\text{g}) - W_{wcc}(\text{g})}{100(\text{cm}^3)} \quad (6)$$

$$\text{Total porosity}(\%) = \left(1 - \frac{\text{Bulk density}(\text{gcm}^{-3})}{\text{Soil density}(\text{gcm}^{-3})}\right) \times 100\% \quad (7)$$

$$\text{Saturated water capacity}(\text{kgkg}^{-1}) = \frac{W_{sat}(\text{g}) - W_{ccs}(\text{g})}{W_{ccs}(\text{g}) - W_{wcc}(\text{g})} \times 100\% \quad (8)$$

$$\text{Capillary porosity}(\%) = \frac{\text{Bulk density}(\text{gcm}^{-3}) \times \text{Capillary holding capacity}(\%)}{\rho_{\text{water}}(\text{gcm}^{-3})} \quad (9)$$

$$\text{Noncapillary porosity}(\%) = \text{Total porosity}(\%) - \text{Capillary porosity}(\%) \quad (10)$$

$$\text{Capillary holding capacity}(\%) = \frac{W_{ws2h}(\text{g}) - W_{ccs}(\text{g})}{W_{ccs}(\text{g}) - W_{wcc}(\text{g})} \times 100\% \quad (11)$$

$$\text{Field capacity}(\%) = \frac{W_{ws5d}(\text{g}) - W_{ccs}(\text{g})}{W_{ccs}(\text{g}) - W_{wcc}(\text{g})} \times 100\% \quad (12)$$

$$x_s(\%) = (1 - \text{Total porosity}(\%)) \times 100\% \quad (13)$$

$$x_l(\%) = \frac{\text{Initial water content}(\text{kgkg}^{-1}) \times \text{Bulk density}(\text{gcm}^{-3})}{\rho_{\text{water}}(\text{gcm}^{-3})} \quad (14)$$

$$x_g(\%) = \text{Total porosity}(\%) - \text{Liquid phase}(\%) \quad (15)$$

$$\text{TPSI} = [(x_s - 25)x_l x_g]^{0.4769} \quad (16)$$

where, TPSI represents the three soil phase index; the higher the values, the better the soil structures. x_s is the volumetric soil content (%), x_l is the volumetric water content (%), and x_g is the volumetric gas content (%) (Wang et al., 2015).

2.6. Statistical analyses

One-way analysis of variance (ANOVA) was conducted to assess the differences in the physical properties between the various soil layers and the different hillslope positions (Lix et al., 1996). ANOVA was also performed to determine the effect of different hillslope positions on the infiltration capacity, preferential flow indices, and root biomass. A two-way multivariate analysis of variance was performed with the soil layer and hillslope position as the main fixed factors. Correlations among the soil hydrological, physical, and root biomass parameters were assessed using the Pearson correlation analysis. Correlation among the various parameters of the soil hydrological and physical characteristics and the root biomass was performed using correlation analysis. Significant correction coefficients were tested at $P < 0.05$, $P < 0.01$, and $P < 0.001$.

3. Results

3.1. Root biomass

The root depth distributions of the total and fine roots in the hilltops, middle slope, and valley bottoms are shown in Fig. 2. The fine and total root biomass were significantly varying with hillslope positions and soil depths. The total and fine root biomass values were significantly higher in the hilltops than in the middle slope and valley bottom ($P < 0.001$) in the shallow soil layer (0–20 cm). The highest total and fine root biomass values occurred in the deep soil layers (20–110 cm); however, the differences were not significant between the hilltops and middle slopes. Across the 0–110 cm range, the total root biomass from the valley bottom (5.77 kg m^{-2}) was significantly higher than that from the hilltops (4.43 kg m^{-2}) and middle slope (3.53 kg m^{-2} ; $P < 0.05$). However, the fine root biomass from the middle slope (2.33 kg m^{-2}) was significantly lower than that from the hilltops (3.33 kg m^{-2}) and valley bottom (3.54 kg m^{-2} ; $P < 0.05$).

3.2. Soil physical properties and structure

All slope positions and soil depths had significant independent and interactive effects on soil properties ($P < 0.001$; Fig. 3). Although the initial gravimetric water content did not significantly differ among the positions at 0–30 cm depths, the values in the valley bottom were significantly lower than those in the hilltops and middle slope at 30–110 cm depths. The large holes created by termites, earthworms, or decayed roots declined significantly from the hilltop to the middle slope and then the valley bottom (Table 1).

The physical properties of noncapillary porosity, total porosity, saturated water capacity, field water capacity, capillary holding capacity, and gas phase were significantly higher, while bulk density was significantly lower in the hilltops than in the middle slope and valley bottom at depths of 0–50 cm (Fig. 3). The decreasing order of bulk density was valley bottom > middle slope > hilltop at 70–110 cm depth.

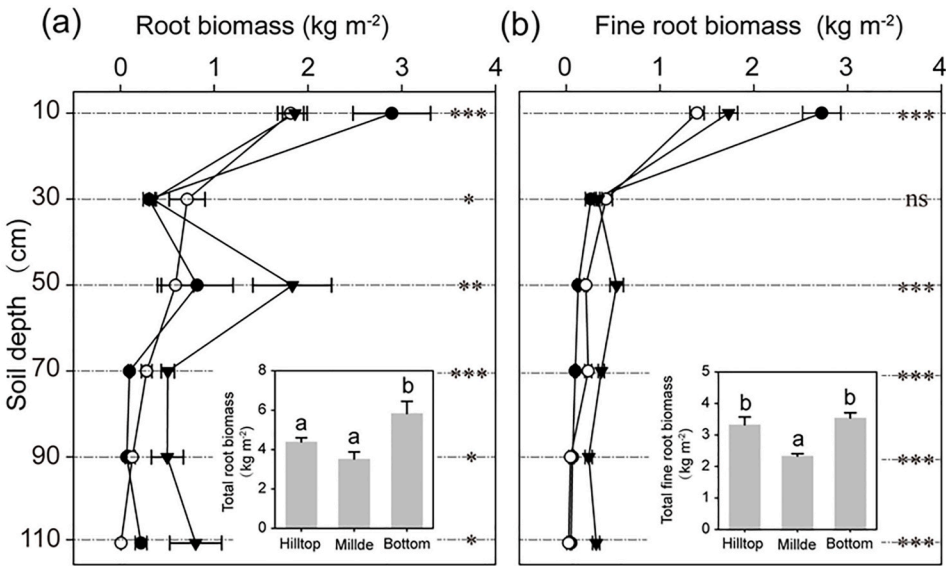


Fig. 2. The distribution of fine root and total root (including coarse and fine roots) biomass across the soil depths at different mountain slopes ($n = 6$). The summation of fine root and total root biomass at different mountain slopes depicted in inset graph ($n = 36$). The black circle represents the hill top, the hollow circle represents the middle slope, and the inverted triangle represents the valley bottom. Hilltop: hill top, Middle: middle slope, Bottom: valley bottom. Values are means $\pm$ SE. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

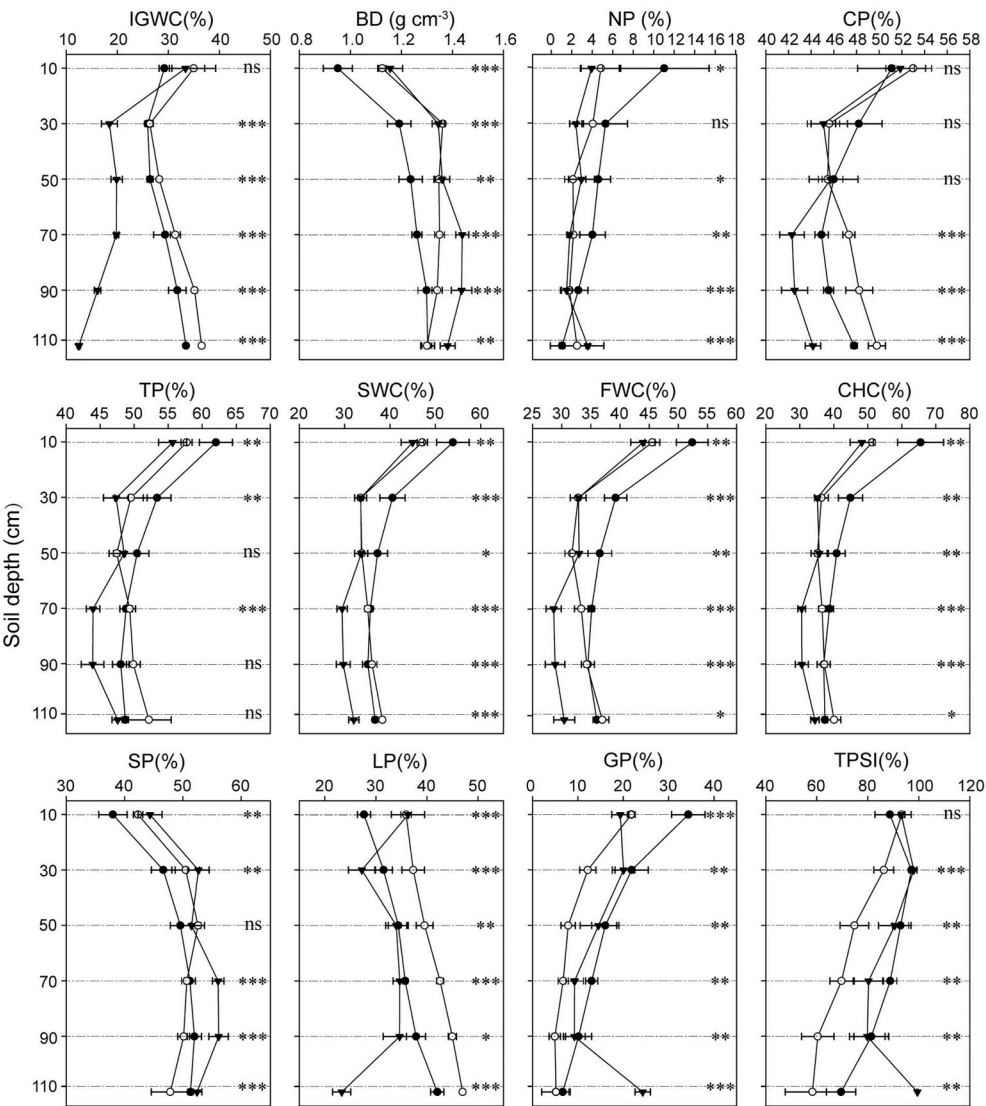


Fig. 3. Physical property differences of each soil layer under the hill top, middle slope, and valley bottom. IGWC: initial gravimetric water content (%); BD: bulk density (g cm⁻³); TP: total porosity (%); CP: capillary porosity (%); NP: noncapillary porosity (%); SWC: saturated water capacity (%); CHC: capillary holding capacity (%); FWC: field water capacity (%); SP: solid phase (%); LP: liquid phase (%); GP: gas phase (%); TPSI: three soil phase index (%). Hilltop: hill top, Middle: middle slope, Bottom: valley bottom. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Table 1
Soil characteristics in the hill top, middle hill, and valley bottom in 0–30 cm.

Item	Hill top	Middle hill	Valley bottom
Percentage of gravel (%)	2.97 ± 0.32 ^a	1.88 ± 0.16 ^b	0 ^c
Number of holes (diameter >1.0 cm)	142.67 ± 5.44 ^a	104.83 ± 7.77 ^b	15.67 ± 1.17 ^c
Holes created by termites or ants	10.83 ± 0.7 ^a	6.67 ± 0.56 ^b	1.33 ± 0.42 ^c
Sand (%)	60.08 ± 1.23 ^a	59.77 ± 1.13 ^a	59.63 ± 0.63 ^a
Silt (%)	24.24 ± 1.68 ^a	23.91 ± 2.04 ^a	24.2 ± 1.53 ^a
Clay (%)	15.68 ± 1.02 ^a	16.32 ± 1.49 ^a	16.17 ± 1.24 ^a

The number of the holes were counted in the sections with 30 cm depth and 100 cm width. Data are expressed as the mean ± standard error ($n = 6$). Different lowercase letters indicate a significant difference at $P < 0.05$.

Furthermore, the capillary holding capacity, total porosity, saturated water capacity, and field water capacity showed almost no significant differences between the middle slope and valley bottom, whereas they were significantly lower in the hilltop at 70–110 cm depth.

3.3. Soil water infiltration rates

The infiltration rates of the hilltop and middle slope locations rapidly decreased over the first 15 min and did not change after 60 min, whereas in the valley bottom these values rapidly decreased for 5 min and did not change after 40 min (Fig. 4). The IIR of the hilltops was 4.88 cm min^{-1} , significantly higher than that of the middle slope (2.25 cm min^{-1}) and valley bottom (1.08 cm min^{-1}). The AIR of the hilltops (1.70 cm min^{-1}) was significantly higher than that of the middle slope ($0.783 \text{ cm min}^{-1}$) and valley bottom (0.22 cm min^{-1}). The K_s values were not significantly different between the hilltops (0.36 cm min^{-1}) and middle slope (0.24 cm min^{-1}); however, these values were significantly higher than those of the valley bottom (0.05 cm min^{-1}).

3.4. Water flow behaviour and preferential flow

The dye-stained water flow patterns were interpreted directly in vertical soil sections of each position (Fig. 5). The dye was distributed uniformly in the soil layers of 0–40 cm in the hilltops and 0–20 cm in the middle slope, while exhibiting random patches in the remaining soil layers. However, the total dye-stained area was uniform across all soil

layers at the valley bottom. The total infiltration time presented a significant decline from the hilltop (4.53 min) to the middle slope (41.33 min) and then the valley bottom (373.33 min). The total dye-stained area (of the excavation surfaces) did not significantly differ between hilltops (3594.87 cm^2) and the middle slope (3769.14 cm^2), while their dye-stained area was higher compared to valley bottom (1769.06 cm^2). The maximum depth of the dye-stained area in the middle slope (112.17 cm) was significantly higher than in the hilltops (95.3 cm), and the latter was significantly higher than that of the valley bottom (49.83 cm; Table 2). There were no significant differences in the maximum width of the dye-stained area between hilltops and middle slope ($P > 0.05$), while both were significantly higher than in the valley bottom ($P > 0.05$; Table 2). Although the dark blue area with high dye concentration ($>2.0 \text{ g L}^{-1}$) was mainly distributed in 0–40 cm of the soil profiles, it was spread to approximately 80 cm depth in the hilltop and 100 cm depth in the middle slope.

All parameters (except for the LI) quantifying the degree of preferential flow showed the same consistent trend: preferential flow was highest in hilltops and lowest in the valley bottom, while the middle slope represented an intermediate stage (Fig. 7a–e). The dye coverage in the hilltops (46.54%) was not significantly different from that in the middle slope (44.80%), but was significantly higher than that in the valley bottom (34.63%; $P < 0.05$). The uniform infiltration depth in hilltops was significantly lower than that in the middle slope ($P < 0.001$), and the latter was significantly lower than in the valley bottom ($P < 0.05$). The preferential flow fraction of the hilltops was significantly higher than that of the middle slope ($P < 0.05$), and the latter was significantly higher than that of the valley bottom ($P < 0.001$). The LI of the hilltops and the middle slope was significantly higher than that of the valley bottom, suggesting that infiltration pathways were more irregular in the hilltops and middle slope and more uniform at the valley bottom.

3.5. Plant roots, soil property, and infiltration parameter correlations

All physical properties were correlated at different levels (Fig. 7a). The IIR, AIR, and K_s had significantly negative correlations with bulk density ($R_{IIR} = -0.81$, $P < 0.05$; $R_{AIR} = -0.70$, $P < 0.05$; $R_{Ks} = -0.73$, $P < 0.01$) and had significantly positive correlations with total root and fine root biomass and total porosity ($P < 0.05$; Fig. 7b). UniFr had a significant positive correlation with bulk density ($R = 0.88$, $P < 0.01$), and significantly negative correlations with capillary holding capacity

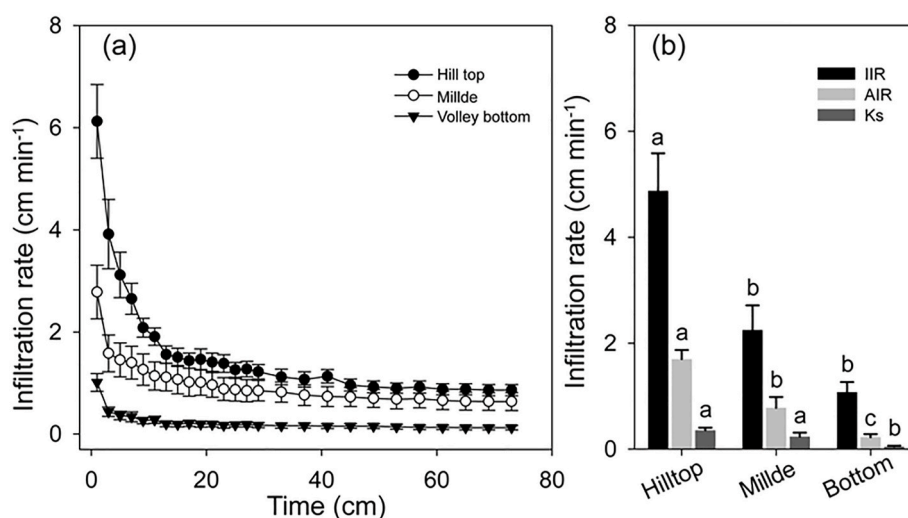


Fig. 4. Variation in the (a) infiltration rate with time (cm min^{-1}) and the (b) initial infiltration rates (IIR, cm min^{-1}), average infiltration rates (AIR, cm min^{-1}) and saturated hydraulic conductivity (K_s , cm min^{-1}) under the hill top, middle slope, and valley bottom. The means and standard error ($n = 8$). Hilltop: hill top, Middle: middle slope, Bottom: valley bottom. Different lowercase letters indicate a significant difference at $P < 0.05$.

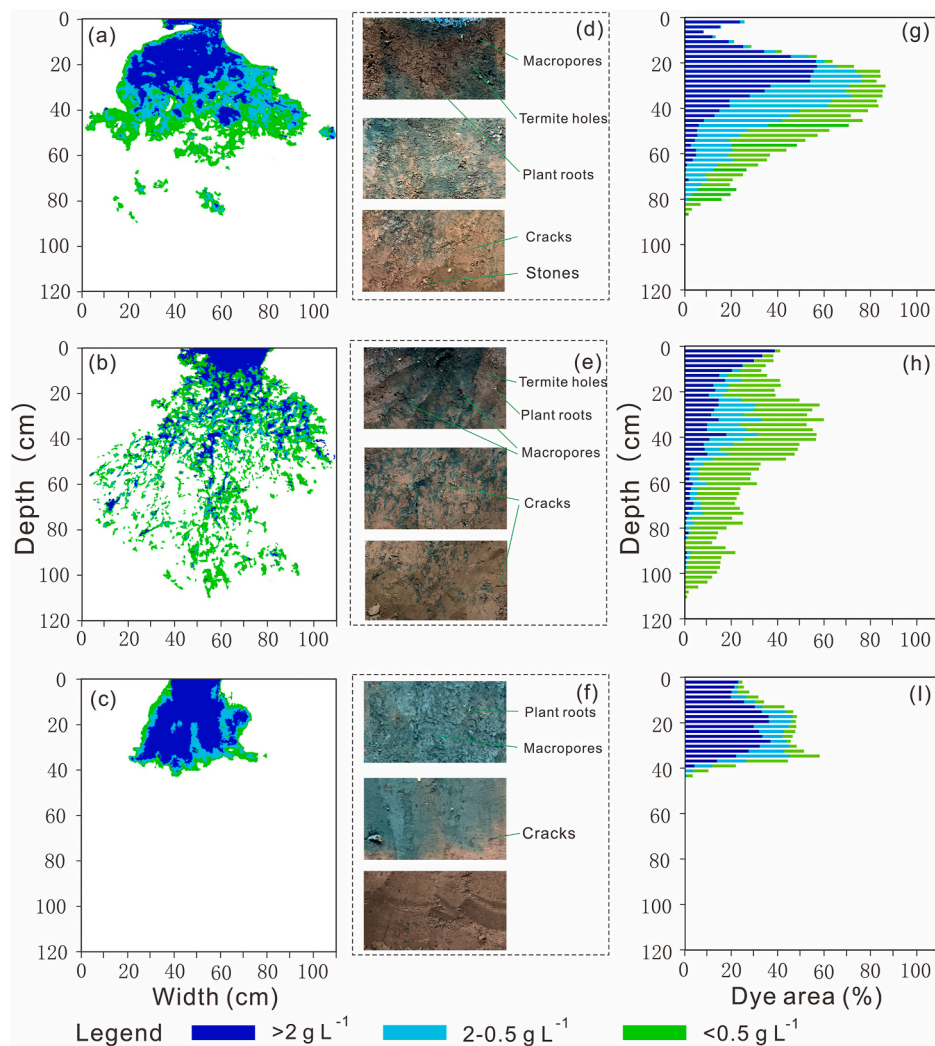


Fig. 5. Dye-stained infiltration patterns for vertical soil sections in (a) hill top, (b) middle slope, and (c) valley bottom. The real structure of different soil layers in (d) hill top, (e) middle slope, and (f) valley bottom. Calculated area (cm²) of the dye-stained soil profile vs. depth (cm) in (g) hill top, (h) middle slope, and (i) valley bottom. Hilltop: hill top, Middle: middle slope, Bottom: valley bottom.

Table 2

Dye stain characteristics in the vertical dye soil profile for hill top, middle hill, and valley bottom.

Treatment	infiltration time (min)	Maximum dye-stained		Total stained area (cm ²)	Area (cm ²) of concentration		
		depth (cm)	width (cm)		>2 g L ⁻¹	2-0.5 g L ⁻¹	<0.5 g L ⁻¹
Hill top	4.93 ^a	95.3 ± 0.66 ^b	104.33 ± 4.10 ^a	3594.87 ± 61.02 ^a	1412.70 ± 195.22 ^a	1102.72 ± 56.49 ^a	1079.44 ± 175.81 ^b
Middle hill	41.33 ^b	112.17 ± 1.10 ^a	105.33 ± 3.71 ^a	3769.14 ± 367.82 ^a	947.60 ± 159.02 ^a	663.17 ± 67.74 ^b	2158.37 ± 175.37 ^a
Valley bottom	373.33 ^c	49.83 ± 5.10 ^c	65.33 ± 1.45 ^b	1769.06 ± 222.12 ^b	1051.79 ± 86.58 ^a	426.01 ± 76.04 ^c	291.26 ± 59.58 ^c

Data are expressed as the mean ± standard error ($n = 3$). Different lowercase letters indicate a significant difference at $P < 0.05$.

($R = -0.86$, $P < 0.01$), field water capacity ($R = -0.88$, $P < 0.01$), saturated water capacity ($R = -0.85$, $P < 0.01$), total root biomass ($R = -0.79$, $P < 0.05$), and fine root biomass ($R = -0.74$, $P < 0.05$). PF-fr had a significant negative correlation with bulk density ($R = -0.84$, $P < 0.01$) and a significantly positive correlation with total root biomass ($R = 0.71$, $P < 0.01$).

4. Discussion

4.1. Vertical variation of infiltration capacity

In the present study, soil infiltration capacities varied with slope positions at different elevations, vegetation coverage, and soil

physicochemical properties. The soil hydraulic conductivities, including IIR, AIR, and Ks, decreased with decreasing slope elevation (from the hilltop to the middle slope and valley bottom). This accords with those results found in previous studies of other climate ecosystems. For example, Zhao et al. (2020) found a K_s increase with increasing slope elevation on a rocky mountain hillslope, attributed to more macropores formed by roots and gravels in the upslope in northern China with temperate climate. Another study also found the same variability in the infiltration capacity on a hillslope in Spanish vineyards in a Mediterranean subtropical climate (Rodrigo-Comino et al., 2016). These results confirm the hypothesis that the hillslope position has a significant impact on the soil infiltration capacity.

Numerous studies have demonstrated that soil structure, surface

roughness, plant roots, organic matter, animal burrows, soil moisture, soil texture, and soil ecogeography are the main factors affecting the infiltration capacity (Morbideilli et al., 2018; Rossi et al., 2018). We found that soil infiltration capacity was significantly affected by the soil physical properties, and that it varied with different slope positions (Figs. 3 and 7, respectively). In both the surface (0–20 cm) and all (0–120 cm) layers, the physical properties deteriorated with decreasing elevation, including reduced total porosity, noncapillary porosity, saturated water capacity, capillary holding capacity, and increase in soil bulk density. The IIR and K_s were significantly correlated with the noncapillary porosity, and IIR and AIR correlated with the total porosity (Fig. 7). Similar results were reported by Jiang et al. (2019) and Tang et al. (2019). They suggested that soils with a favourable structure may have higher porosity, allowing water to infiltrate into the deep soil layers. These results indicate that the spatial variation in water infiltration capacity at different hillslope positions can be attributed to differences in soil physical properties.

Previous studies have found that root channel indices, including mean diameter and area, positively influence the initial and average infiltration rates (Wu et al., 2017; Guo et al., 2019). In addition, total and fine root biomass have been correlated with initial and average infiltration rates (Cui et al., 2019), which we also found in our study (Fig. 7). The slope angle is another factor contributing to the infiltration rate. In a laboratory simulation, Khan et al. (2016) demonstrated that infiltration rate decreased with increasing slope gradient and increased with rainfall intensity. Furthermore, holes created by ants and termites contribute to the formation of soil macropores, therefore leading to the vertical variation in infiltration capacity (Bargués Tobella et al., 2014; Chen et al., 2019, Table 1).

4.2. Preferential flow and its contributing factors

Dye tracers reveal the degree and pattern of preferential flow and offer visual information regarding water movement that bypasses the soil matrix and pore spaces (Allaire et al., 2009). Preferential flow indices (Fig. 6), infiltration time, and dye stain patterns (Table 2) showed a clear trend of decline from hilltops to middle slope and valley bottom. Additionally, higher *PI* and *LI* values in the top and middle hillslopes suggest that the water flowed uniformly through the soil matrix down to deeper depths, in contrast to the valley bottom, where the preferential flow patterns were uniform. This result agrees with Zhao et al. (2020), who found that the degree of preferential water flow and the spatial non-uniformity of infiltration distribution increased with

increasing hillslope elevation in the Taihang Mountains in a temperate climate in China. Wang et al. (2018) also found that preferential flow, coupled with the downslope infiltration capacity, was lower on the lower slopes than on the upper slopes in karst regions. By contrast, Mei et al. (2018) found the opposite results. They found that abundant rock fragments in the downslope locations resulted in higher preferential flow and infiltration variability in semiarid loess hillslopes. These results indicate that not only hillslope position but also other factors, such as soil type or properties, could also affect preferential flow behaviours.

Rock fragments, root systems, faunal activities, and soil properties in different geographic locations are the main factors that affect preferential water flow paths in hillslopes (Zhu et al., 2014; Mei et al., 2018). Previous studies demonstrated that cracks created by rock fragments can increase preferential pathways (Zhang et al., 2016; Qi et al., 2020). We also found similar results. The hilltop and middle slope are characterised by limestone soils with numerous stones, while the valley bottom is a zone of mountain and river sediments without stones (Table 1). In addition, the channels formed by both living and decayed roots are the main preferential channels that provide continuous routes for macropore flow (Bargués Tobella et al., 2014). Typically, decayed roots in the soils of the valley bottom form macropores in the soil layer. However, macropores capture fine sediments from upslope and the river in the valley bottom, resulting in a relatively uniform structure, low levels of macropore network, and reduced preferential flow (Hanson et al., 2004; Zwartendijk et al., 2017). In addition, higher earthworm and termite activities in the top and middle slope soils could produce more macropores (i.e., channels and burrows), promoting water movement through preferential pathways (Van Schaik et al., 2014; Cheng et al., 2017). Furthermore, a lower bulk density, higher soil porosity and moisture, and numerous fissures in soils also increased preferential flow paths in the top and middle hillslopes (Fig. 3).

Preferential flow strongly affects the water infiltration capacity and increases the complexity and messiness in the redistribution pattern of water in soils (Alaoui et al., 2011). In our study, the dyed water infiltration time significantly increased with decreasing elevation from hilltop to middle slope and valley bottom. This indicates that the hilltop has a higher potential to infiltrate the rainfall during the rainy season, whereas the valley bottom has a poor drainage discharge capacity. Moreover, the preferential water flow parameters (*DC*, *UniFr*, and *Tot-StAr*) were positively correlated with the infiltration capacity (IIR, AIR, and K_s), suggesting the preferential water paths could facilitate the movement of water in soil profiles and accelerate water redistribution in the hillslope.

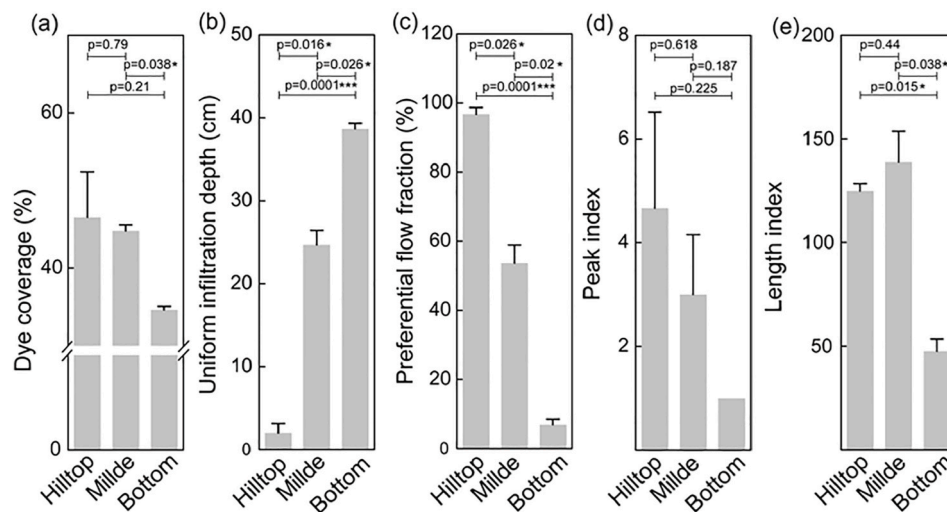


Fig. 6. The different measured parameters of preferential water flow (a) dye coverage, (b) uniform infiltration depth, (c) preferential flow fraction, (d) length index, and (e) peak index) in hill top, middle slope, and valley bottom. *P* values for the Wilcoxon sign ranked test are also shown.

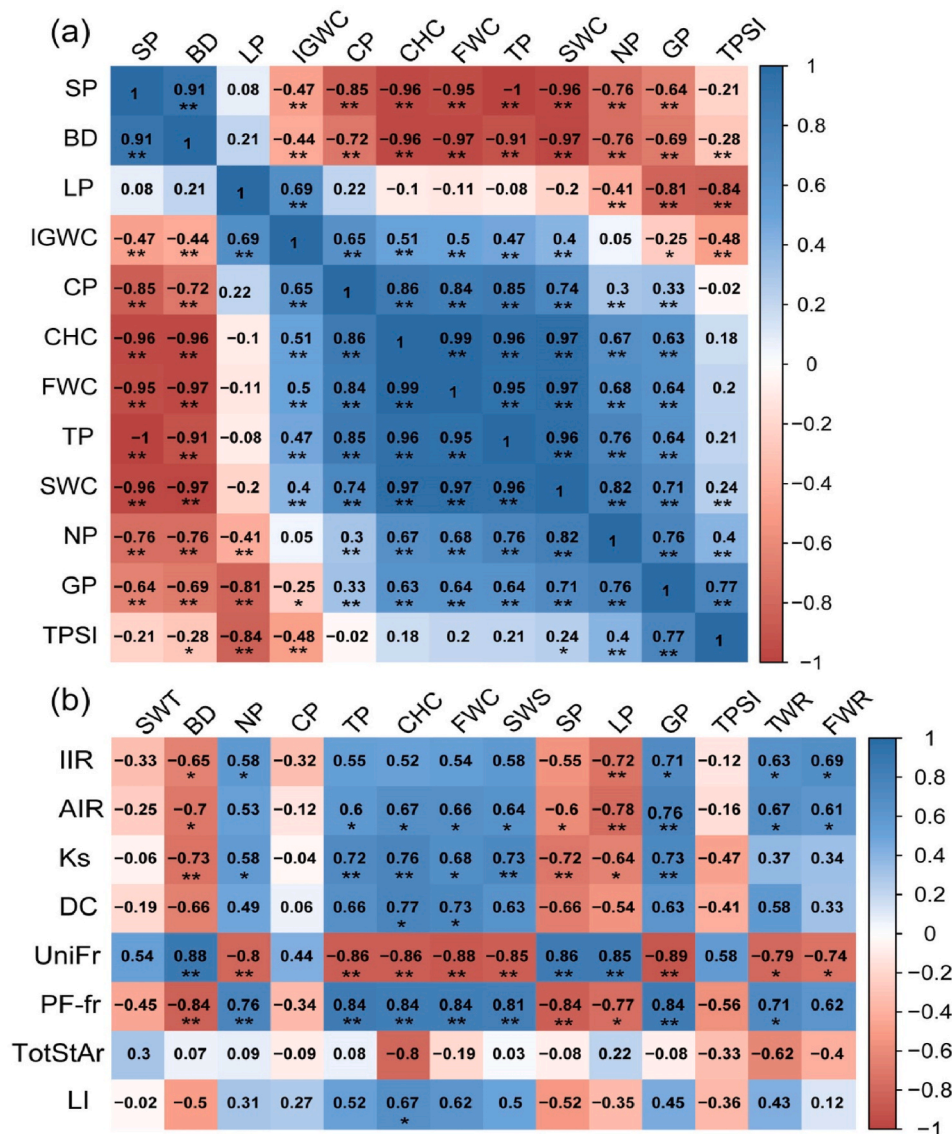


Fig. 7. Correlation coefficients among (a) soil physical properties and (b) among soil water infiltration capacity, preferential water flow, and root biomass parameters as well as physical properties. The blue colour represents positive correlations, and the red colour represents the negative correlations. The colour intensity of the blocks is proportional to the correlation coefficients. In the right side of the correlogram, the legend colour shows the correlation coefficients and the corresponding colours. IGWC: initial gravimetric water content (%); BD: bulk density (g cm^{-3}); TP: total porosity (%); CP: capillary porosity (%); NP: noncapillary porosity (%); SWC: saturated water capacity (%); CHC: capillary holding capacity (%); FWC: field water capacity (%); SP: solid phase (%); LP: liquid phase (%); GP: gas phase (%); TPSI: three soil phase index (%); AIR: average infiltration rate; IIR: initial infiltration rate; FWR: fine root biomass; TWR: total root biomass. ID: Infiltration depth (cm), DC: Dye coverage (%), UniFr: Uniform infiltration depth (cm), TotStAr: Stained area (cm^2), PF-fr: Preferential flow fraction (%), LI: Length index. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Improved preferential flow and water exchange likely contribute to the flow and redistribution of soil water in the vertical and horizontal directions, thereby diversifying water sources and supply (Zhu et al., 2019). In the hilltop and middle slope, the active water zone extended down to the 80–100 cm depth layer, while at the valley bottom it was concentrated in the shallow layer (Fig. 5). Furthermore, large amounts of the water buffer zone were found around the preferential flow paths, indicating that the soils in the upper hillslope locations could accelerate water exchange between the soil profiles. These results suggest that the trees can efficiently use water from both the shallow and deep layers in the upper locations of the hill slope during the dry season. Conversely, low preferential water flow paths and water exchange between soil profiles in the valley bottom require longer roots to absorb and utilise deep soil water, thereby promoting interspecific water competition (Fig. 2).

4.3. Implications for hydrological management and potential limitation

In Xishuangbanna, high rainfall variability, accompanied by excessive rainfall and frequent drought events due to climate change, lead to narrow streamflow and water storage capacity in the dry season and accelerated surface runoff and soil erosion in the rainy season at both the

hillslope and watershed scales (Li et al., 2016). This series of environmental and hydrological problems is closely related to soil infiltration ability and water movement paths. Whether rainfall infiltrates into the soil or flows along the surface depends largely on the magnitude of surface K_s relative to the prevailing rainfall intensity (Zwartendijk et al., 2017). Unlike the valley bottom, the relatively high K_s values in the hilltops exceed most rainfall intensities observed in this zone (Figs. 1 and 4), indicating that infiltration-excess overland flow may hardly occur in the upper slopes but frequently occurs in the lower slopes. Thus, rainwater in the upper hillslopes can quickly percolate the vertical soil profile and supplement the groundwater, or flow through the soil's preferential paths to the streams, contributing to deep drainage and groundwater recharge (Fig. 8). The replenishment of shallower and deeper groundwater resources could increase the soil water content in the upper hillslopes compared to the valley bottom, with higher evaporation losses and lower soil moisture (Fig. 3). Moreover, enhanced overland runoff—characterised by over-saturation in the lower hillslopes—probably results in soil erosion, land degradation, and subsequent nutrient depletion at the valley bottom. Therefore, more attention should be given to improving water exchange and soil fertility, reducing soil erosion, and enhancing soil productivity during watershed management in the valley bottom. In addition, the model of

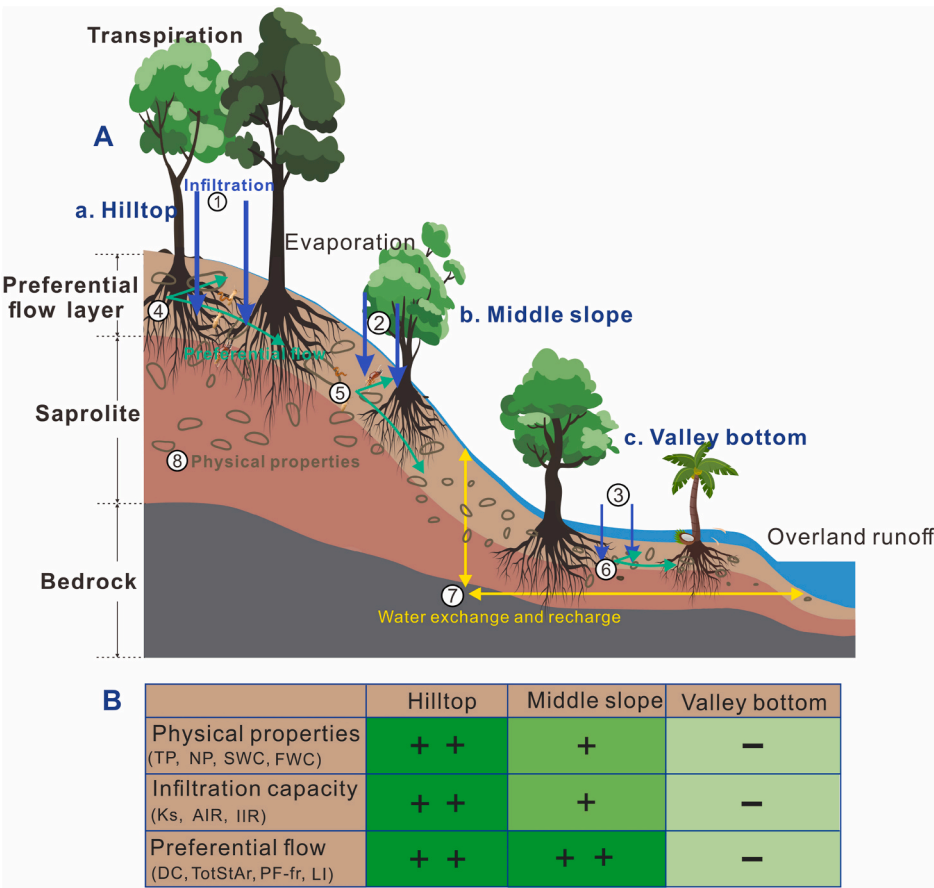


Fig. 8. Schematic diagram illustrating the effects of hillslopes on the hydrological processes in the rainy season (A), and inserted table summarising changes in physical properties, infiltration capacity, and preferential flow among hilltop, middle slope, and valley bottom (B). The infiltration and preferential water flow in the upper locations were significantly higher than that in the valley bottom due to differences in the water content, soil physical properties, root systems, animal activities, cracks and stones. “++” indicates better status, “+” refers to moderate status, “-” indicates deterioration. See Fig. 7 for abbreviations.

saturated-unsaturated infiltration should be developed in the tropical forest hillslopes to simulate infiltration into heterogeneous soils from the soil unsaturated to saturated conditions during stormwater (Rahardjo et al., 2019).

In addition, water infiltration capacity and preferential flow are closely related to hydraulic redistribution and water supply mechanisms in soil profiles, thereby determining plant water use efficiency (Jiang et al., 2017; Zhu et al., 2019). In general, trees prefer to acquire water—and build more roots—in soil layers with relatively high water content (Ma and Song, 2016). In the present study, water collected in soils during the rainy season can be redistributed rapidly through the preferential flow paths in the upper hillslopes, resulting in a higher water content in each soil layer (Fig. 3). However, in the valley bottom, the lower water content in the soil layers would force the roots to penetrate deeper soil layers searching for water (Figs. 2 and 3). These results are supported by those of Schwendenmann et al. (2015) and Wang et al. (2019). In a stable isotope study, it was found that some tree species prefer to use shallow soil water in the dry season, and some take up water mainly from deeper layers, but most tree species get water uniformly from the upper and deeper layers in the tropical mountainous region (Schwendenmann et al., 2015). However, in the valley bottom, most trees primarily use deep soil water (below 50 cm), primarily from the rivers during the dry season and to a smaller extent from rainfall (Wang et al., 2019). These studies indicate that water resources available to plants are controlled by hydrological processes, including water infiltration and preferential flow. However, current studies referred to the combination between water processes in soils (e.g., movement, redistribution, and storage of water) and the water use strategy of the plants is still limited (Sprenger et al., 2015). Therefore, further research on the distribution characteristics of water resources and how this affects plant water acquisition is needed at both the hillslope and

watershed scales.

5. Conclusions

Our study examined the effects of hillslope position on hydrological functions, including water distribution, infiltration capacity (IIR, K_s , and AIR), preferential water flow, and water exchange. Infiltration capacity, including IIR, K_s , and AIR, decreased from the hilltop to the middle slope and the valley bottom. The results suggested that the infiltration-excess overland flow, and consequently excessive soil erosion, may hardly occur in the upper slopes but commonly occurs in the lower slopes. Preferential flow, and its spatial heterogeneity, was significantly higher in the upper (top and middle) slopes than in the valley bottom. The infiltration and preferential flow parameters were significantly correlated with soil water content, noncapillary and total porosity, root biomass, and holes created by termites. These results indicated that the soil properties—coupled with the root systems, animal activities, cracks and stones—remarkably affected the soil infiltration capacity and preferential flow. Furthermore, differences in soil hydraulic properties between the different hillslope positions contributed to the redistribution, transportation, and storage of surface and belowground water. Our data can be helpful to understand hydrological processes in tropical mountain regions concerning soil water management for plants and be used as a reference for eco-hydrological studies in tropical ecosystems.

Credit author statement

Chunfeng Chen: Conceptualization, Funding acquisition, Methodology, Investigation, Formal analysis, Visualization, Writing – original draft. Xin Zou: Data curation, Formal analysis. Ashutosh Kumar Singh: Writing – review & editing. Xiai Zhu: Investigation. Wanjuan Zhang:

Investigation. Bin Yang: Writing – review & editing. Xiaojin Jiang: Formal analysis, Writing – review & editing. Wenjie Liu: Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2021.113672>.

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